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A group of paleobotanists collected samples of sediments containing pollen and spores from various plant species. Each sample was analyzed to determine the number of spores in the sample, the number of pollen grains in the sample, and the species to which those spores and pollen grains belong. Samples S1 and S2 were collected and analyzed in this way. For these two samples:

S1 had the greater total number of spores, and it contained pollen grains from Species P as well as spores from Species Q.

S2 had the greater total number of species represented, and all of the spores it contained were from Species Q.

Statement: Among these two samples, S1 had the greater ratio of total number of ____1____ to total number of ____2____.

Select for *1* and for *2* the options that complete the statement so that it is accurate based on the information provided. Make only two selections, one in each column.

1	2	
		pollen grains
		pollen grains from Species P
		species represented
		spores
		spores from Species Q

Questions 142 to 170 — Difficulty: Hard

142. Biologist: Conservation biologists working to prevent species extinction have long acknowledged that popular species such as lions, eagles, and pandas receive disproportionately large amounts of funding and public attention as compared to less-popular species such as invertebrates and amphibians. Indeed, many of these less-popular species are more in danger of extinction than the more-popular species. Although many conservation biologists have accepted this pattern of disproportionate funding, I believe it needs to stop. For, despite the substantial and continuing expenditure of resources on the more-popular species, very few of these species have any chance of escaping extinction.

The biologist's reasoning is subject to the criticism that the claim that ____A____, which is used to justify the main point, undermines the support for the point that ____B____.

Select for *A* and for *B* the options such that criticism of the zoologist's reasoning is strongest. Make only two selections, one in each column.

A	B	
		popular species receive a disproportionate amount of the money and public attention devoted to the preservation of endangered species
		funding and public attention should not be wasted on the preservation of endangered species
		certain popular species are more endangered than many believe
		many species that are not popular are likely to escape extinction
		very few of the more-popular species have any chance of escaping extinction

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143. Historian: In the collection of the Science Museum, London, there is a small bottle that is purported to contain the entire original batch of artificial mauve dye created by Sir William Perkin in his laboratory in 1856. Indeed, in his experiments that preceded his commercial production of mauve dyes, he made only a few grams of the substance, but this batch would likely have been completely used up when he tested it as a dye. Also, the early version of mauve dye that Perkin produced commercially consisted of an impure paste rather than the pure crystalline form of the substance contained in the bottle.

Statement: The historian's statements imply that the bottle does not contain ____1____ at least in part because the substance in the bottle is ____2____.

Select for *1* and for *2* the options that create the statement that is most strongly supported by the information above. Make only two selections, one in each column.

1	**2**	
		any of Perkin's original batch of mauve dye
		a dye produced in Perkin's laboratory
		a dye created in 1856
		a commercially produced dye
		a crystalline form of the dye

144. When a user conducts a search of a certain library database, the database returns the results as a list of items (books, magazines, etc.) that match the search criteria. The user can select from two settings to establish a maximum number of items—either 12 or 24—to be displayed at a time. If the list contains more items than the established maximum, the list will be displayed on multiple pages, with each page (with the possible exception of the last page) containing the established maximum number of items and with each item in the list appearing on exactly one of the pages. The maximum established by the user does not affect the total number of items listed.

For the list of n items returned by a particular search of the database, the number of items, X, displayed on the last page when the established maximum is 12 is different from the number of items, Y, displayed on the last page when the established maximum is 24.

Select for X and for Y values that are jointly consistent with the information provided. Make only two selections, one in each column.

X	**Y**	
		1
		5
		7
		12
		17

145. In a group of 100 tourists visiting Scandinavia, 10% visited Norway, Sweden, and Denmark; 48% visited only two of these countries; 42% visited only one of these countries; and 100% visited at least one of these countries. Twenty-three tourists in the group visited at least Norway and Sweden, and 28 tourists in the group visited both Norway and Denmark. Furthermore, 3 tourists in the group visited only Denmark, and 32 tourists in the group visited only Sweden.

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visited only Sweden.

In the table, select the number of tourists in the group who *visited at least Norway*, and select the number of tourists in the group who *visited at least Sweden*. Make only two selections, one in each column.

Visited at least Norway	Visited at least Sweden	
		40
		48
		54
		62
		68
		72

146. A city council member standing for reelection has been asked to rate how much she agrees or disagrees that various cultural establishments should receive more city funding. Expressing her overall views regarding budgeting priorities, she answers "Absolutely agree" for the art museum, "Somewhat agree" for the children's theater, "Neither agree nor disagree" for the historical society, "Somewhat disagree" for the symphony orchestra, and "Absolutely disagree" for the metropolitan ballet.

Suppose that the city council member's responses reflect genuine opinions that are logically consistent with her other opinions, and suppose that she believes that funding for any one of the institutions does not affect funding for any of the others. In the first column of the table, rate how much she most clearly agrees that *At least two* of the five institutions should receive more funding. In the second column, rate how much she most clearly agrees that *At most four* of the five institutions should receive more funding. Make only two selections, one in each column.

At least two	At most four	
		Absolutely agrees
		Somewhat agrees
		Neither agrees nor disagrees
		Somewhat disagrees
		Absolutely disagrees

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147. There are exactly three steps—Steps 1, 2, and 3, in that order—used to assemble light bulbs at a particular factory. Some of the bulbs fail to make it through every step in the assembly process; if a bulb fails a given step, it will not proceed to the next step. For the most recent month, the factory manager knows the number of bulbs that were started as well as each of the following:

A = the number of bulbs making it through Step 1 only;

B = the number of bulbs making it through at least Step 2;

C = the number of bulbs making it through all three steps.

The manager wishes to know the fraction of the bulbs that made it through Step 1 that also made it through all three steps. In the table, choose for *Numerator* and *Denominator* the expressions that will give the factory owner the fraction desired. Make only two selections, one in each column.

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Numerator	Denominator	
		A
		B
		C
		A + B
		A − C
		B + C

148. Legal advocate: The Métis people of Canada are of First Nations and European ancestry. The government grants certain special rights to Métis individuals. To receive these rights, an individual must self-identify as Métis and must not have self-identified only recently in order to receive these rights. The individual must also have Métis ancestry and be accepted as Métis by a modern Métis community. Acceptance by a modern Métis community is shown only by long-standing participation in the community's cultural or political activities.

In the table, select characteristics H and N, such that an individual having H and NOT having N would most clearly satisfy the legal advocate's stated criterion for receiving the rights associated with Métis membership. Make only two selections, one in each column.

H	N	
		Self-identifies as Métis due to long-standing Métis ancestry
		Self-identifies as Métis due to long-standing participation in Métis political activities
		Has recent ancestry that is Métis and participates in Métis political activities
		Has recent ancestry that is not Métis
		Has never participated in Métis political activities

149. Tickets for next month's production of *The Sea Gull* at the local community theater went on sale yesterday. The ticket price for a person older than 12 is $\$A$, and the ticket price for a person 12 or under is $\$C$, where $A > C$. Yesterday, the sale of 33 tickets generated a total revenue of \$323, and today the sale of 44 tickets generated a total revenue of \$424.

In the table, select a value for A and a value for C that are jointly consistent with the given information. Make only two selections, one in each column.

A	C	
		6
		8
		10
		11

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		12

150. Companies A and B are part of the same industry and are located in the same city. For Company A, the average (arithmetic mean) salary of its employees, in United Arab Emirates dirhams (AED), is 10,000 AED higher than that for Company B. However, more than half of the employees at Company A have salaries below the average for Company B.

Statement: If the average salary at Company B is ____1____, then the median salary at Company A is ____2____.

Select for *1* and for *2* the options that complete the statement so that it most accurately reflects the information provided. Make only two selections, one in each column.

1	2	
		greater than 100,000 AED
		less than 100,000 AED
		equal to 110,000 AED
		between 100,000 and 110,000 AED
		greater than 110,000 AED

151. In comparing the results of a recent annual spring census to those of the previous year, biologists observed dramatic changes in the numbers of frogs of two species—Species X and Species Y—and an overall decrease in the combined number of frogs of these species. The biologists hypothesized that this decrease was caused by the unusually cold weather between the two censuses.

Select for *A* and for *B* the statements such that the biologists' hypothesis would have the most support if A is true and B is false. Make only two selections, one in each column.

A	B	
		Species X is susceptible to unusually cold weather, whereas Species Y is not.
		Species X has a later mating season than does Species Y.
		Both Species X and Species Y are susceptible to similar contaminants.
		The number of Species X frogs decreased from the previous spring census.
		The number of Species Y frogs decreased from the previous spring census.

152. A certain company defines its *annual labor turnover rate* as the number of employees who left the company during the year divided by the average of the number of employees on the first day of the year and the number of employees on the last day of that year. Last year, the company had a labor turnover rate of exactly 25%.

Assuming that the information above is true, select for *Statement 1* and for *Statement 2* the statements such that if Statement 1 is true, then Statement 2 must be true, but it could be the case that Statement 2 is true and Statement 1 is false. Make only two selections, one in each column.

Statement 1	Statement 2	

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		Last year, more people left the company than began work at the company.
		Exactly 25% of the employees working at the company on the first day of last year left the company last year.
		More than 25% of the employees working at the company on the first day of last year left the company last year.
		All and only those who were employees at the company on the first day of last year were employees on the last day of last year.
		Last year, the number of employees on the first day of the year was equal to the number of employees on the last day of the year.

153. Newspaper editor: Published photojournalism must always present the events covered without distorting those events. Sometimes an image distorts the events a photojournalist attempts to capture, such as when the photojournalist has an equipment failure. In such cases, photojournalists are permitted to make minimal changes to an image, but only to the extent that (1) they are certain about what they observed, and (2) those changes reduce the distortion. Photojournalists' photos are often compelling works of art as well as documents of newsworthy events. However, for publication as journalism, a photo's aesthetic features are allowed to be considered only after the photo is shown to accurately portray relevant features of the event it depicts. In accordance with these criteria, in our journalistic coverage of Event E, we elected to publish photograph P1 rather than photograph P2.

Statement: Assuming the editor's statements are true, if ____1____, then it must be the case that ____2____.

Select for *1* and for *2* the two different options that create the statement that is most strongly supported by the information provided. Make only two selections, one in each column.

1	2	
		P1 was altered
		P2 was altered
		P1 and P2 were taken with malfunctioning equipment
		P1 was chosen over P2 at least in part for its aesthetic features
		both P1 and P2 accurately portray the relevant features of Event E

154. A stock trader working for a hedge fund has estimated that the stock of a certain company has probability 0.4 of increasing in price by at least 5 dollars during a certain trading day and probability 0.1 of increasing in price by at least 10 dollars during the trading day.

Based on the trader's estimates, select for X and for Y the options such that the following statement is most accurate. Make only two selections, one in each column.

If the trader multiplies ____X____ by the reciprocal of ____Y____, then the result is the probability that the price of the stock will increase by at least 10 dollars during the trading day, given that the price increases by at least 5 dollars during the trading day.

X	Y	
		0.1
		0.4

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		0.4
		0.1 divided by 0.4
		the reciprocal of 0.1
		the product of 0.1 and 0.4

155. Throughout a certain decade in a European city, the mean monthly rents for studio apartments varied yearly, from a low of €804 to a high of €1,173. Those for one-bedroom apartments also varied yearly, from a low of €1,060 to a high of €1,497. But some individual studio apartments rented for as little as €420 in some years, and some one-bedroom apartments rented for up to €2,262. To visually assess how the ratio of mean monthly rents for studio apartments to those for one-bedroom apartments varied yearly over the decade, Maria requires a graph with the following characteristics. The graph will have two axes of equal length, with mean rents for one-bedroom apartments shown on the horizontal axis and mean rents for studio apartments shown on the vertical axis, and with the same scale on both axes. For each year, mean rents will be plotted as a point.

From the following options, select for *Horizontal axis* a range for the points on the horizontal axis and select for *Vertical axis* a range for the points on the vertical axis that together would satisfy Maria's requirements for the graph. Make only two selections, one in each column.

Horizontal axis	**Vertical axis**	
		€0 to €1,500
		€400 to €1,100
		€800 to €1,200
		€1,000 to €2,300
		€1,100 to €1,500

156. A statistician reached the following conclusions about games between university soccer teams: Overall, a team playing on its home field has a 45% chance of a win, a 25% chance of a loss, and a 30% chance of a draw (a tied outcome). In the games where one or more goals are scored, the team that scores the first goal has a 55% chance of scoring it in the game's first half and a 45% chance of scoring it later in the game. When that team is the home team (i.e., a team playing on its home field), there is a 40% chance that the other team will score no goals at all, and therefore a 60% chance that it will score one or more goals.

Select for X and for Y two different outcomes such that the information provided explicitly includes the statistician's estimates of the probability that if X occurs, so will Y. Make only two selections, one in each column.

X	**Y**	
		The home team scores at least two goals in the game.
		The home team scores the first goal.
		A goal is scored in the first half of the game.
		A goal is scored in the second half of the game.
		The team opposing the home team scores at least one goal.

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157. To ensure computer security, a firm has rules about access of managerial and nonmanagerial employees to various networked computer drives. In accordance with the rules, some but not all nonmanagerial employees are allowed access to both Drive B and Drive D. Some are required to have such access. Every nonmanagerial employee must have access to at least one of Drive B and Drive D. No nonmanagerial employee is allowed to have access to both Drive B and Drive E. The rules do not restrict the access of managerial employees to any of the drives mentioned.

From the following statements about computer security practices in relation to the firm's employees, select for *Required* the statement that, based on the information provided, describes a practice that is required by the rules, and select for *Permitted* the statement that describes a practice that is permitted, but NOT required, by the rules. Make only two selections, one in each column.

Required	**Permitted**	
		At least one nonmanagerial employee is currently not allowed access to any of the three drives B, D, and E.
		Managerial employees, and only those, are currently allowed access to Drive E.
		The nonmanagerial employees who are currently allowed access to Drive D are also allowed access to Drive E.
		The managerial and nonmanagerial employees who are currently allowed access to Drive B are also allowed access to Drive E.
		Any nonmanagerial employees currently allowed access to Drive E are also allowed access to Drive D.

158. Giulia is planning to sell her car, which is fueled by gasoline (petrol) and averages 20 miles per gallon (mpg), and purchase a diesel-fueled car that averages 30 mpg. She estimates that her future cost per gallon of diesel fuel will be 5% higher than her present cost per gallon of gasoline. She wishes to estimate (1) the annual cost of fuel for her new car if she maintains her present annual total miles driven and (2) the annual total miles she can drive her new car if she maintains her present annual expenditure on fuel.

Let x represent Giulia's present annual cost per gallon of gasoline in US dollars, and let y equal her present annual total of miles driven. Select for *Cost* an appropriate expression for Giulia's estimate of (1) above, and select for *Miles* an appropriate expression for her estimate of (2) above. Make only two selections, one in each column.

Cost	**Miles**	
		$\frac{2}{3}(1.05x)$
		$\frac{3}{2}\left(\frac{y}{1.05}\right)$
		$\frac{3}{2}(1.05y)$
		$\frac{1.05xy}{30}$
		$\frac{1.05xy}{}$

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		$\frac{\quad}{20}$
		$\frac{20xy}{1.05}$

159. Each of three botanists made a hypothesis regarding specimens of a particular plant species:

Botanist 1: Any individual specimen possessing the gene for curly stems has either the gene for long roots or the gene for purple flowers, or both.

Botanist 2: Any individual specimen possessing the gene for long roots has either the gene for flat leaves or the gene for round seeds, or both.

Botanist 3: No individual specimen that possesses either the gene for curly stems or the gene for flat leaves or both has the gene for purple flowers.

The discovery of an individual specimen of the plant species in question having the gene for ____1____ but NOT the gene for ____2____ would show that at least one of the three hypotheses described is incorrect.

Select for *1* and for *2* the characteristics that would most accurately complete the statement, based on the information given. Make only two selections, one in each column.

1	2	
		curly stems
		flat leaves
		long roots
		purple flowers
		round seeds

160. An inventory of a neighborhood's trees found that 32 percent were conifers and most of the rest were deciduous. Among the conifers were 258 spruces and 112 pines, along with some cedars and other species. Most of the deciduous trees were oaks, but one in eight was a maple. Of the oaks, 65 percent were red oaks and 25 percent were white oaks. Of the maples, 20 percent were Japanese maples.

Select for A and for B two types of trees such that the ratio of the number of trees of the type selected for A to the number of trees of the type selected for B can be determined and is less than 1. Make only two selections, one in each column.

A	B	
		cedars
		conifers
		deciduous trees
		Japanese maples
		red oaks
		spruces

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161. A manufacturing company plans to begin automating production and reducing its workforce by installing industrial robots at the start of each year over five years. Each robot will result in annual labor-cost savings of €150,000 beginning with the year of its installation and will have an up-front cost of €400,000. For robot installation in Year 1 of the program, the company will budget €600,000, plus the total amount of labor costs that will be saved by robotic production in that year. The robot-installation budget for each subsequent year will consist of the total amount of labor costs that will be saved by robotic production in that year plus any money left over from the previous year's robot-installation budget.

Statement: Within budget constraints, the maximum number of robots that can be installed in Year 1 of the program is ____X____, and the maximum number of robots that can be in service in Year 3 is ____Y____.

Select values for X and for Y that create the statement that follows logically from the information provided. Make only two selections, one in each column.

X	Y	
		2
		3
		4
		5
		6
		7

162. While designing a game involving chance, Desmond noticed that the probability a fair coin lands face up exactly 2 times when the coin is tossed 3 times is equal to the probability that a fair coin lands face up exactly m times when the coin is tossed n times, where $n > 3$.

Select for m and for n values consistent with the given information. Make only two selections, one in each column.

m	n	
		1
		2
		3
		4
		5
		6

163. Linguist: Plosives and fricatives are two classes of consonants. A "voicing contrast" is a distinction between two consonants that are identical except that one is voiced and the other is unvoiced. In language family X, languages with voicing contrasts in their fricatives always have voicing contrasts in their plosives. This means that in that family, any given language has a voicing contrast in its fricatives _____ it has a voicing contrast in its plosives. In other words, a given language in that family lacks any voicing contrasts in its plosives _____ it lacks any such contrasts in its fricatives.

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Select for *First blank* the word or phrase that most logically completes the statement with the first blank. And select for *Second blank* the word or phrase that most logically completes the statement with the second blank. Make only two selections, one in each column.

First blank	**Second blank**	
		and
		if
		only if
		or
		unless

164. At a certain university, there is a strong positive correlation between the time of day at which university classes are offered and the classes' average (arithmetic mean) grades, with earlier times associated with lower grades. However, for the classes offered at the earliest time but not for any other classes, there was a strong negative correlation between the grades of students in those classes and the number of scheduled classes they missed. In fact, when the grades of students who missed at least 5 scheduled classes were excluded, classes offered at the earliest time more often than not had significantly higher average grades than classes offered at any other time.

Consider the following statement:

At the university in question, classes offered at earlier times ____1____ lower average grades than classes offered later in the day, but when the grades of students who missed at least 5 scheduled classes were excluded, classes offered at the earliest time____2____ higher average grades than classes offered later in the day.

Select for *1* and for *2* the options that complete the sentence so that it most accurately summarizes the information provided. Make only two selections, one in each column.

1	**2**	
		tended to have
		tended not to have
		almost always had
		seldom, if ever, had
		possibly had

165. Each Monday through Friday (a *workweek*), Avinash will bring either exactly one apple or exactly one banana with him to his workplace for an afternoon snack. To avoid having to decide which to bring, each morning Avinash will toss a coin with a face on exactly one side that is equally likely to land face up or face down. If the coin lands face up, then he will bring an apple, and if the coin lands face down, then he will bring a banana. Avinash correctly determined the probability that, for a given workweek, either he would bring an apple on at least 4 consecutive days or he would bring a banana on at least 4 consecutive days. This probability was m divided by n.

Select for m and for n values jointly consistent with the given information. Make only two selections, one in each column.

m	n	

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		4
		6
		8
		12
		20
		32

166. Philosophy student: Some objects that are considered beautiful by everyone who has observed them may not be, in fact, truly beautiful. To see that this is so, consider this: No one doubts that some objects that are appreciated by many people have aesthetic flaws that are discernible only to sophisticated observers. But even these sophisticated observers are limited by their finite intellects and experiences. Thus, an object that appears beautiful to the most sophisticated actual observers may nonetheless have subtle but severe aesthetic shortcomings that would make it appear hideous to hypothetical observers of even greater sophistication. Such an object would be ugly, regardless of any actual person's opinion.

In general, if an object ____1____, then that object ____2____.

Select for *1* and for *2* the two different options that complete the sentence in such a way that it expresses a principle on which the philosophy student's argument relies. Make only two selections, one in each column.

1	**2**	
		is considered beautiful by everyone
		is thought by most observers to have some aesthetic flaws
		would appear hideous to hypothetical observers of even greater sophistication than the most sophisticated actual observers
		is not truly beautiful
		is not widely appreciated by unsophisticated observers

167. The following statements describe certain characteristics of a certain pool of candidates for a position. Any candidate who did not meet the minimum qualifications for the position was immediately excluded from consideration. The two candidates who met the minimum qualifications for the position and met all of the desired qualifications also had multiple recommendations. All candidates who received a telephone interview also had extensive experience. All candidates who had extensive experience and impressed the hiring committee during the telephone interview were invited to interview on-site. At least one candidate declined an invitation for an on-site interview, and exactly one candidate was interviewed on-site without receiving a telephone interview.

Consider the following incomplete sentence:

If any candidate ____1____, then that candidate ____2____.

Select for *1* and for *2* two different options that best complete the sentence such that it can be logically inferred from the information provided. Make only two selections, one in each column.

1	**2**	
		did not meet the minimum qualifications

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		had multiple recommendations
		had extensive experience
		impressed the hiring committee during the telephone interview
		interviewed on-site

168. On a 12-hour analog clock, the hour hand moves at a constant rate of 1 revolution every 12 hours, and the minute hand moves at a constant rate of 1 revolution every hour. The hands are perpendicular at 3:00 in the morning. To the nearest second, the next time they are superimposed (i.e., both pointing at the same point on the outer rim of the clock face) is M minutes and S seconds after 3:00 in the morning.

Select for M and for S values that are consistent with the information provided. Make only two selections, one in each column.

M	S	
		15
		16
		17
		20
		22
		34

169. The following argument is logically flawed. The author's goal was to craft the argument so that the conclusion follows logically from Premises 1 and 2 and so that both premises are necessary to draw the conclusion.

Premise 1: Every respondent to our survey who **reported feeling satisfied** also reported being in a good mood.

Premise 2: Every respondent to our survey who **reported having a central goal** also **reported being in a good mood**.

Conclusion: Therefore, assuming all of the reports were accurate and complete, every respondent to our survey who **felt satisfied** also **had a central goal**.

Select for *Boldface A* and for *Boldface B* two of the boldfaced phrases in the argument such that Boldface A occurs earlier in the argument than Boldface B, and exchanging the positions of those two phrases in the argument would make it so the argument fulfills the author's goal. Make only two selections, one in each column.

Boldface A	**Boldface B**	
		reported feeling satisfied
		reported having a central goal
		reported being in a good mood
		felt satisfied

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		had a central goal

170. In Country C, some but not all eligible voters are required to vote. The particulars of the country's laws governing voting are as follows:

Every citizen who is eligible must vote on election day.

A person is eligible if (and only if) he or she meets the age requirement and either is a citizen or meets the residency requirement for noncitizens.

The age requirement is that every voter must be at least 19 years old on election day.

The residency requirement for noncitizens is that the voter must have been a resident of Country C for at least 5 years on election day.

Consider the following individuals:

Abigail: a citizen who is currently 19 years old

Barbara: a 7-year resident noncitizen who is currently 19 years old

Charles: a 7-year resident noncitizen who is currently 18 years old

For an election held today, select the individual or individuals who must vote, based on the information provided, and select the individual or individuals who must not vote, based on the information provided. Make only two selections, one in each column.

Must vote	**Must not vote**	
		Abigail only
		Barbara only
		Charles only
		Abigail and Barbara only
		Abigail and Charles only
		Barbara and Charles only

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